

Online Appendix

Warehouses and Diplomas: Amazon's Expansion and Local Educational Attainment

Md Shafiqul Islam

This Online Appendix reports results referenced in the main paper. Section [OA1](#) reports additional attainment results, Section [OA2](#) additional enrollment results and the timing of the attainment and enrollment responses, Section [OA3](#) the event-study estimates for household income and the labor market, and Section [OA4](#) synthetic difference-in-differences estimates and the estimates from the IPUMS microdata panel.

OA1 Additional Attainment Results

OA1.1 Heterogeneity by Gender and Race

Figure [OA1](#) and Table [OA1](#) report the effects separately for men and women. The bachelor's share rises by 0.48 percentage points among men and 0.48 among women, and the master's share by 0.49 and 0.30; the gains are broad-based across genders. Figure [OA2](#) and Table [OA2](#) report the bachelor's effect by race and ethnicity. The effect among White adults, 0.47 percentage points, is precisely estimated and close to the overall effect. The estimates for Black, Hispanic, and Asian adults rest on the few commuting zones (53, 47, and 11) in which these estimates are published for every county, and they are imprecise. All estimates use the same sample and specifications as Table [2](#).

Table OA1: Effects of Amazon's Entry on Educational Attainment by Gender

	BA+ Men (1)	BA+ Women (2)	MA+ Men (3)	MA+ Women (4)
Amazon Entry	0.477** (0.204)	0.475** (0.186)	0.492*** (0.117)	0.298** (0.118)
Mean Dep. Var.	28.17	27.84	10.61	10.00
SD	6.77	6.71	3.48	3.31
Communing zones	188	188	188	188
N	3,572	3,572	3,572	3,572

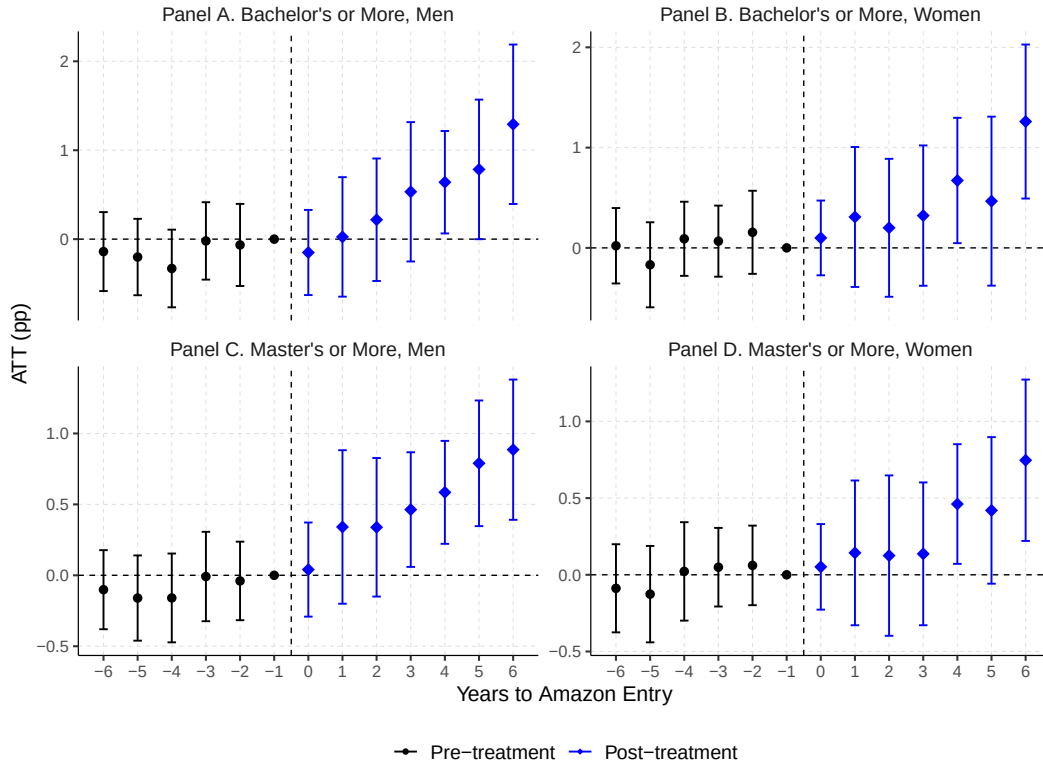
Notes: Dependent variable is the percentage-point share of men or women aged 25 and older with at least a bachelor's degree (columns 1–2) or a master's degree (columns 3–4), from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates are the doubly-robust staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#), averaged over the first seven years after treatment (event years 0–6), comparing treated commuting zones to not-yet-treated controls. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024; a commuting zone is treated from the year before its first facility opens (anticipation). Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. and SD report the pre-treatment mean and standard deviation in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table OA2: Effects of Amazon's Entry on Bachelor's Attainment by Race and Ethnicity

	White (1)	Black (2)	Hispanic (3)	Asian (4)
Amazon Entry	0.470** (0.203)	0.789 (0.675)	1.183* (0.649)	2.847** (1.331)
Mean Dep. Var.	31.33	16.70	13.20	40.81
SD	7.67	5.86	6.85	11.72
Communing zones	187	53	47	11
N	3,553	1,007	893	209

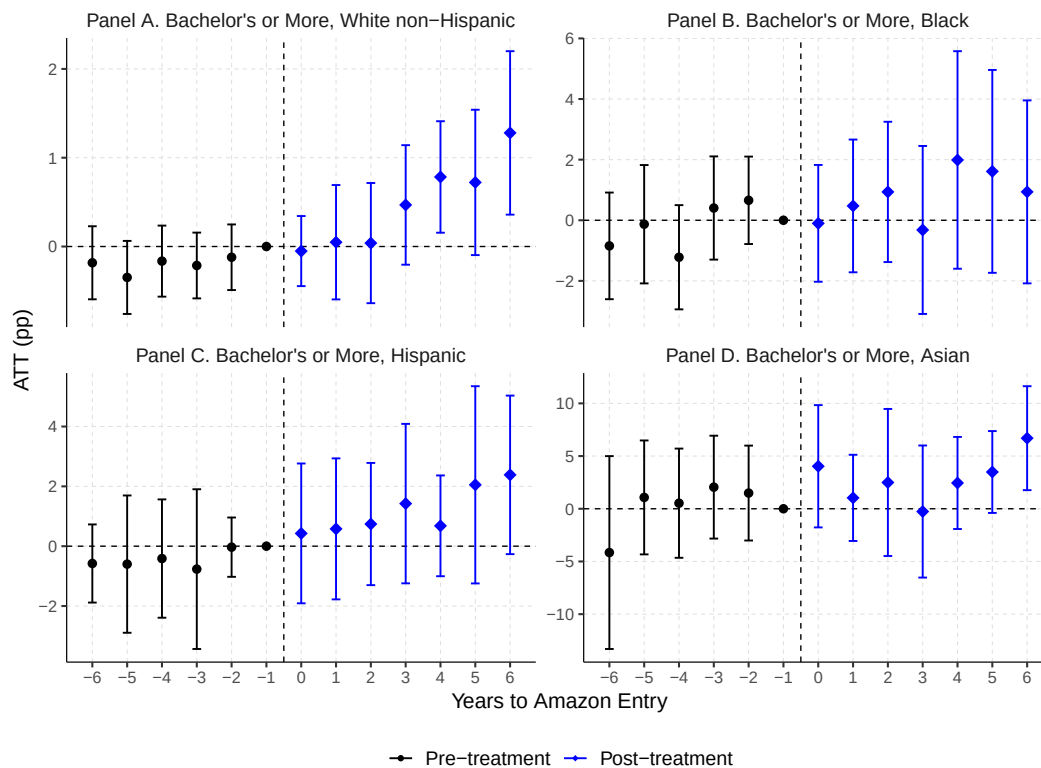
Notes: Dependent variable is the percentage-point share of adults aged 25 and older in the specified racial or ethnic group with at least a bachelor's degree, from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Estimates are the doubly-robust staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#), averaged over the first seven years after treatment (event years 0–6), comparing treated commuting zones to not-yet-treated controls. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024; a commuting zone is treated from the year before its first facility opens (anticipation). For each column, the sample excludes commuting zones in which any county's estimate for the group is suppressed by the Census Bureau in any year. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. and SD report the pre-treatment mean and standard deviation in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure OA1: Event-Study Estimates by Gender, 2005–2024



Notes: Each panel plots event-time coefficients; black circles are pre-treatment and blue diamonds post-treatment coefficients. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon's first facility opens in the commuting zone (to allow for anticipation). Dependent variable is the percentage-point share of men or women aged 25 and older with at least a bachelor's degree (Panels A–B) or a master's degree (Panels C–D), from published American Community Survey 1-year estimates for counties with at least 65,000 residents (2005–2024; 2020 is not available), aggregated to 2020 commuting zones. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant'Anna \(2021\)](#), comparing treated commuting zones to not-yet-treated controls.

Figure OA2: Event-Study Estimates of Bachelor's Attainment by Race and Ethnicity, 2005–2024



Notes: Same as Figure OA1, except that the dependent variable is the percentage-point share of adults aged 25 and older in the specified racial or ethnic group with at least a bachelor's degree. For each group, commuting zones in which the Census Bureau suppresses any county's estimate for the group in any year are excluded. Estimates for Black, Hispanic, and Asian adults rest on few commuting zones and are correspondingly imprecise.

OA2 Additional Enrollment Results

OA2.1 Enrollment by Gender and Race

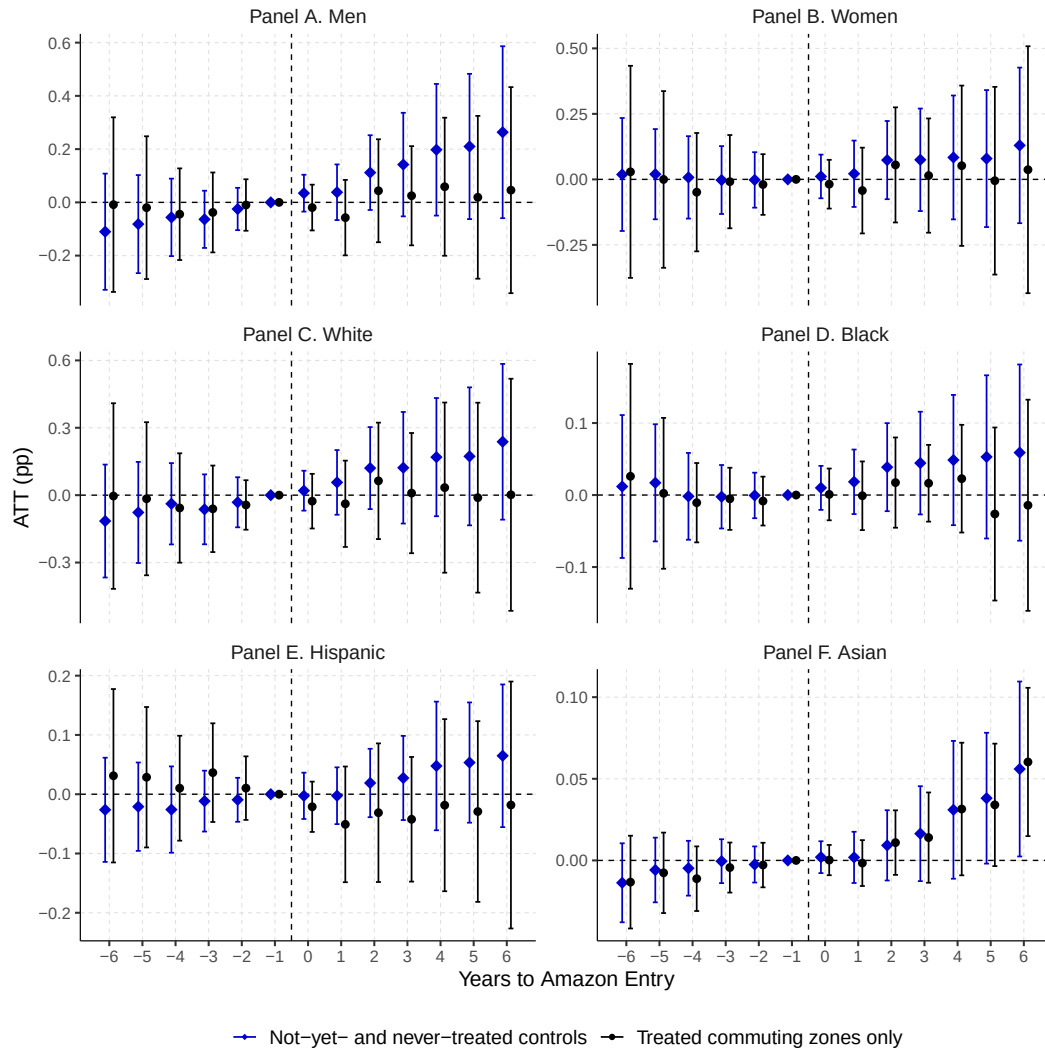
IPEDS reports first-time enrollment by gender and race for each institution, so I re-estimate the four-year results separately by group, using the same local-draw colleges, the same treatment definition, and the commuting zone's total population aged 18–24 as the denominator for every group. Figure OA3 and Table OA3 report the estimates. The increase in four-year first-time enrollment is concentrated among men: 0.144 percentage points (standard error 0.065, $p = 0.03$), about 6 percent of the pre-treatment mean, compared with 0.070 (standard error 0.060) for women. First-time enrollment of Asian students rises by 0.020 percentage points ($p = 0.03$), about 12 percent of a small pre-treatment mean of 0.16; it is the only group estimate that is also statistically significant when the sample is restricted to treated commuting zones (0.015, $p = 0.02$). Estimates for White and Black students are positive and marginally significant (0.124 and 0.043), and the estimate for Hispanic students is not significant. At two-year colleges, first-time enrollment of Hispanic students falls by -0.070 percentage points ($p = 0.02$), while other groups show no change. Because these estimates span twelve group-by-college-type outcomes, isolated significant coefficients should be read with caution; the gender pattern and the Asian estimate, which holds in both samples, are the most robust.

OA2.2 Online Enrollment

If Amazon's entry draws young adults toward traditional four-year study, it may also change how students attend. Since fall 2012, IPEDS has reported the number of each institution's students who take all of their courses through distance education. I aggregate these counts to commuting zones in the same way as first-time enrollment and measure the exclusively online share as the percentage of all students at local-draw public four-year or public two-year colleges who study entirely online. Restricting to local-draw public colleges matters more here than for first-time enrollment: large online programs enroll students nationwide but report them at the institution's home campus, so an online share measured at all colleges partly reflects where national online providers are located rather than how local residents enroll.

The sample ends in 2019. In spring 2020 the COVID-19 pandemic forced colleges to move nearly all instruction online, and many kept extensive online teaching through 2021 and beyond: nationally, the exclusively online share rose from 17 percent in fall 2019 to 45 percent in fall 2020 and was still 26 percent in 2023. These changes had nothing to do with Amazon's entry, but their size and timing differed across places, with larger and more urban commuting zones, which are also more likely to be treated, returning to in-person instruction more slowly. Including 2020–2023

Figure OA3: Event-Study Estimates: First-Time Four-Year Enrollment by Gender and Race, 2010–2023



Notes: Each panel plots event-time coefficients for first-time degree- or certificate-seeking undergraduate enrollment of the specified group at local-draw four-year colleges, as a percentage of the commuting zone’s total population aged 18–24. Blue diamonds use not-yet-treated and never-treated commuting zones as comparisons; black circles restrict the sample to commuting zones that receive an Amazon facility. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. Asian includes Pacific Islander. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant’Anna \(2021\)](#).

Table OA3: First-Time Enrollment at Local-Draw Four-Year Colleges by Gender and Race

	Men (1)	Women (2)	White (3)	Black (4)	Hispanic (5)	Asian (6)
<i>Panel A. Not-yet- and never-treated controls</i>						
Amazon Entry	0.144** (0.065)	0.070 (0.060)	0.124* (0.075)	0.043* (0.025)	0.030 (0.024)	0.020** (0.009)
Commuting zones	325	325	325	325	325	325
<i>Panel B. Treated commuting zones only, not-yet-treated controls</i>						
Amazon Entry	-0.001 (0.084)	-0.001 (0.090)	0.000 (0.122)	0.003 (0.019)	-0.038 (0.060)	0.015** (0.007)
Commuting zones	187	187	187	187	187	187
Mean Dep. Var.	2.28	2.80	3.21	0.60	0.58	0.16

Notes: Dependent variable is first-time degree- or certificate-seeking undergraduate enrollment of the specified group at local-draw four-year colleges in the commuting zone, as a percentage of the commuting zone's total population aged 18–24, 2010–2023. Group counts are from IPEDS Fall Enrollment; Asian includes Pacific Islander in all years for a consistent series. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. A commuting zone is treated from the year before its first Amazon facility opens. Estimates are the doubly-robust staggered difference-in-differences ATT of Callaway and Sant'Anna (2021), aggregated across treatment cohorts; the estimator differences out time-invariant commuting-zone characteristics and common year shocks. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

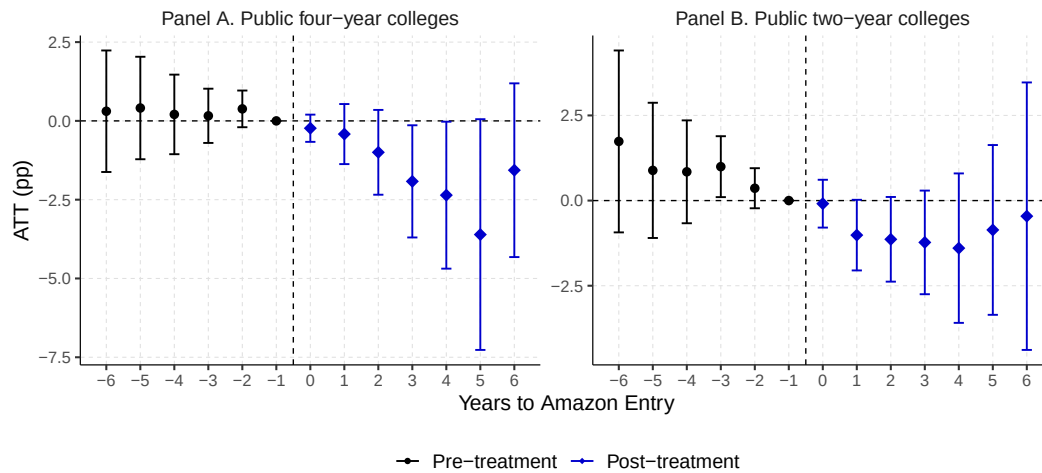
would confound the treatment effect with differences in pandemic-era instruction, so this analysis uses 2012–2019 only. The shorter window also means that commuting zones treated in 2012 or earlier have no pre-period and do not contribute.

Table OA4 reports the estimates, and Figure OA4 plots them. Before the pandemic, the exclusively online share falls after Amazon's entry at both types of college: by -0.700 percentage points at public four-year colleges (standard error 0.334, $p = 0.04$), about 7 percent of the pre-treatment mean of 10.6, and by -0.671 percentage points at public two-year colleges (standard error 0.319, $p = 0.04$). At four-year colleges the decline grows to -2.36 percentage points four years after entry. Together with the fall in the part-time share, this suggests that students shift toward full-time, in-person attendance. The two-year estimate should be read with caution: its pre-treatment coefficients are positive and decline toward zero, so part of the post-treatment decline may continue an earlier trend.

OA2.3 Why Attainment Rises Before Enrollment Can Contribute

The timing puzzle. In the IPUMS microdata panel (Section OA4), the share of adults aged 25 and older with a bachelor's degree rises by 0.47 percentage points within two years of Amazon's entry and by about 1.0 percentage points after four years; in the main panel, the rise is more gradual (Section 4.2). A bachelor's degree usually takes four or more years of full-time study, and the

Figure OA4: Event-Study Estimates: Exclusively Online Share of Enrollment, 2012–2019



Notes: Each panel plots event-time coefficients for the percentage of all students at local-draw public colleges of the specified type in the commuting zone who take all of their courses through distance education. Black circles are pre-treatment and blue diamonds post-treatment coefficients. The sample is 2012–2019: distance-education enrollment is first reported in IPEDS in fall 2012, and the pandemic years 2020–2023 are excluded because COVID-19 forced colleges to move instruction online for reasons unrelated to Amazon. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon’s first facility opens in the commuting zone. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant’Anna \(2021\)](#), comparing treated commuting zones with not-yet-treated and never-treated commuting zones.

enrollment responses in Section 4.4 start small and build gradually. Students who enroll after Amazon arrives cannot graduate in time to explain the early increase in attainment, so the question is which margin moves first.

How students enroll. The first clue is in how students enroll. In commuting zones that receive an Amazon facility, the part-time share of enrollment falls by 1.62 percentage points, and the decline widens to 3.7 percentage points after six years (Table 4). Table OA5 counts full-time and part-time students separately and shows that this decline combines two different responses. Local-draw public four-year colleges grow through full-time students: full-time enrollment rises by 8.1 percent, while part-time enrollment does not change. Local-draw public two-year colleges lose students: part-time enrollment falls by 10.9 percent and full-time enrollment by 4.1 percent. Before the pandemic, the share of students studying entirely online at local public colleges also fell (Table OA4). Amazon’s entry therefore shifts local enrollment toward full-time, in-person study at four-year institutions, and draws adults who combine work and study out of community colleges. The second response is consistent with a tighter local labor market raising the opportunity cost of part-time study, as [Charles et al. \(2018\)](#) document for the housing boom.

Table OA4: Exclusively Online Share of Enrollment at Local-Draw Public Colleges, 2012–2019

	Public four-year (1)	Public two-year (2)
Amazon Entry	–0.700** (0.334)	–0.671** (0.319)
Mean Dep. Var.	10.60	11.72
Commuting zones	262	367
Treated commuting zones	145	158

Notes: Dependent variable is the percentage of all students enrolled at local-draw public colleges of the specified type in the commuting zone who take all of their courses through distance education, 2012–2019. Distance-education enrollment is from the IPEDS Fall Enrollment distance-education component, first collected in fall 2012. The sample ends in 2019 because the COVID-19 pandemic moved nearly all instruction online in 2020 and 2021 for reasons unrelated to Amazon’s entry; nationally, the exclusively online share rose from 17 percent in 2019 to 45 percent in 2020. Four-year colleges are bachelor’s-granting institutions and two-year colleges do not award bachelor’s degrees. Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. A commuting zone is treated from the year before its first Amazon facility opens; cohorts treated in 2012 or earlier have no pre-period and are excluded. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant’Anna \(2021\)](#), aggregated across treatment cohorts, using not-yet-treated and never-treated commuting zones as comparisons; the estimator differences out time-invariant commuting-zone characteristics and common year shocks. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. is the pre-treatment mean in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The enrollment response is too small to explain early attainment. Across all colleges, full-time enrollment rises by 0.19 students per 100 adults aged 25 and older two years after entry (Table OA5, Panel B). This is less than half of the increase in the bachelor’s share at the same horizon, even if every additional student were 25 or older and graduated at once. In practice, most full-time students are younger than 25, and few graduate within two years. The shift toward full-time study is a pipeline: it can raise attainment in later years, but it cannot produce the early increase.

What moves first: the stock of educated adults. The early increase comes instead from the existing adult population. In the main data, the number of adults with a bachelor’s degree grows by 2.6 percent, about twice as fast as the number without one, which grows by 1.4 percent (Table 3). In the microdata, the bachelor’s share rises by 0.80 percentage points among adults who did not move in the previous year, and the share of recent movers does not change. Among adults who moved from another state or from abroad in the past year, the number with a bachelor’s degree rises by about 7 percent, while the number without one does not change (Table 3). Together, these patterns suggest that Amazon’s arrival quickly raises local demand for college-educated workers, which helps communities keep and attract degree holders, while the education of local young people responds more slowly. Because the ACS records only moves within the past year, it cannot separate newcomers who arrived earlier from graduates who would otherwise have left; both are changes in

who lives in the commuting zone rather than new degrees earned locally.

Two margins, two speeds. The attainment and enrollment results therefore describe two margins that move at different speeds. The stock of educated adults responds within a few years of entry. The flow into college shifts more slowly, away from part-time and online study and toward full-time four-year enrollment, and it should add to attainment only as these cohorts graduate, largely beyond the sample period. Two caveats apply. In the microdata without covariates, the pre-treatment attainment coefficients rise before entry (from -0.31 six years before), and with baseline covariates the overall bachelor's estimate is smaller (0.32 , $p = 0.10$). The size of the early stock response is therefore less certain than its direction.

Table OA5: Full-Time and Part-Time Enrollment in Treated Commuting Zones, 2010–2023

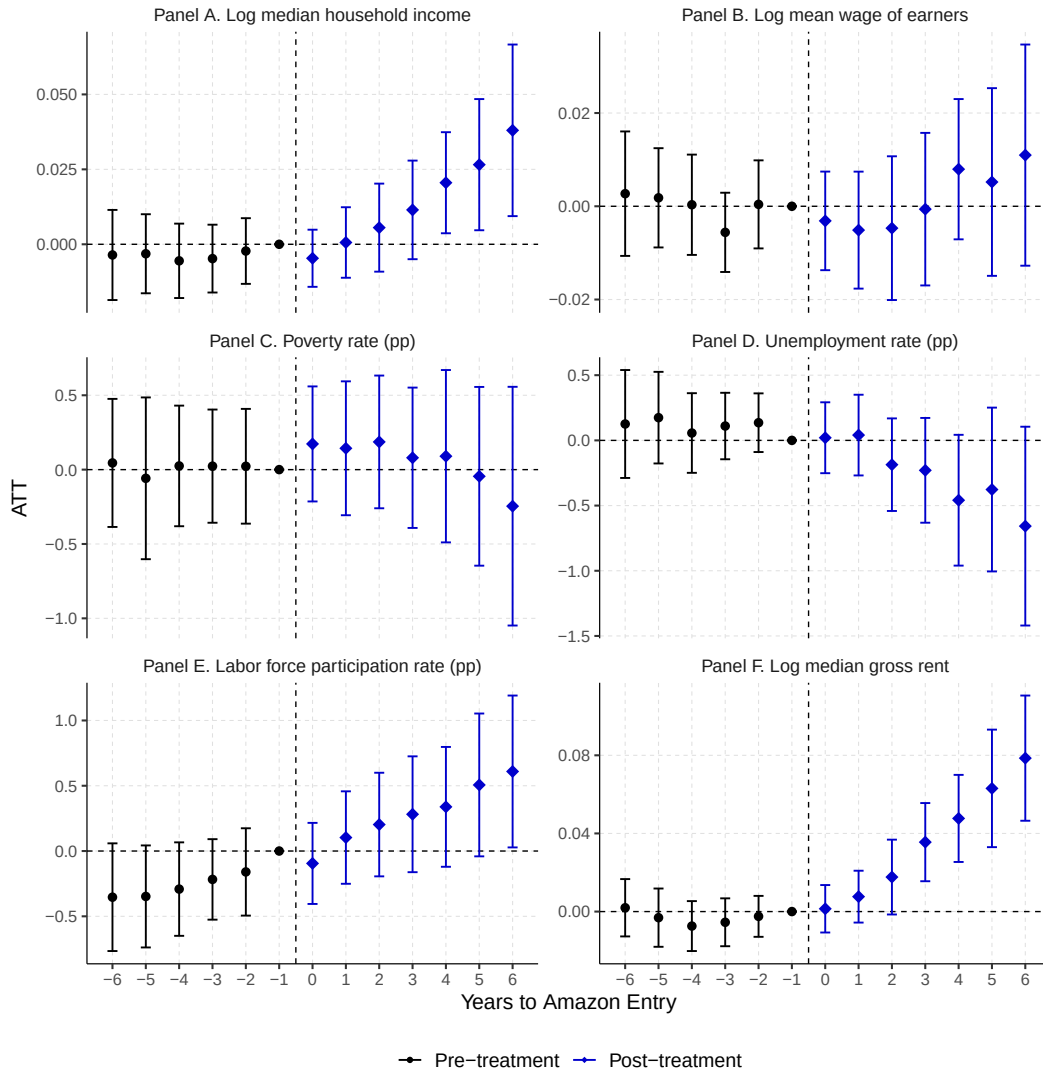
	All colleges (1)	Public four-year (local-draw) (2)	Public two-year (local-draw) (3)
<i>Panel A. Log number of students</i>			
Full-time students	0.046** (0.018)	0.078*** (0.023)	−0.041* (0.025)
Pre-treatment mean	19,993	11,519	3,778
Part-time students	−0.008 (0.032)	0.027 (0.041)	−0.116*** (0.034)
Pre-treatment mean	10,903	4,109	4,883
<i>Panel B. Students per 100 adults aged 25 and older</i>			
Full-time students	0.199* (0.112)	0.201*** (0.068)	0.008 (0.019)
Pre-treatment mean	7.02	4.14	1.29
Part-time students	−0.101 (0.167)	−0.013 (0.056)	−0.104** (0.041)
Pre-treatment mean	3.73	1.47	1.83
Commuting zones	192	159	170

Notes: Each cell is a separate estimate of the effect of Amazon's entry on full-time or part-time enrollment (all students, undergraduate and graduate) at colleges of the specified type in the commuting zone, 2010–2023, from IPEDS Fall Enrollment. Panel A uses the log number of students; Panel B divides the number of students by the commuting zone's population aged 25 and older from IPUMS ACS microdata, the population in which educational attainment is measured. Pre-treatment means are in levels (Panel A: students per commuting zone, geometric mean). Local-draw colleges exclude institutions whose share of first-time students from other states or abroad was 30 percent or more in 2006 and 2008. The sample is restricted to commuting zones that receive an Amazon facility, so commuting zones treated later serve as comparisons for those treated earlier; a commuting zone is treated from the year before its first Amazon facility opens. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant'Anna \(2021\)](#), aggregated across treatment cohorts. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

OA3 Household Income and the Labor Market

Figure OA5 reports the event-study estimates behind Appendix Table A7 of the main paper: household income, the mean wage, poverty, unemployment, labor force participation, and rents.

Figure OA5: Event-Study Estimates: Effects of Amazon's Entry on Household Income, Employment, and Rents, 2005–2024



Notes: Each panel plots event-time coefficients for the specified outcome, from IPUMS ACS 1-year microdata aggregated to 2020 commuting zones, 2005–2024. Panels A, B, and F are in logs of constant 1999 dollars; Panels C–E are in percentage points. Black circles are pre-treatment and blue diamonds post-treatment coefficients. The sample is the balanced panel of commuting zones that receive an Amazon facility, with not-yet-treated commuting zones as comparisons. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon's first facility opens in the commuting zone. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant'Anna \(2021\)](#).

OA4 Educational Attainment: Synthetic DiD and Microdata Estimates

This section re-estimates the attainment effects of Section 4.2 with a second estimator and a second data set.

Synthetic difference-in-differences. The synthetic difference-in-differences estimator of [Arkhangelsky et al. \(2021\)](#), adapted to staggered adoption, builds its comparison group differently from the [Callaway and Sant’Anna \(2021\)](#) estimator. For each treatment cohort, it reweights commuting zones that never receive an Amazon facility so that their pre-treatment path matches that of the treated commuting zones, and it weights pre-treatment years so that they resemble the post-treatment period. The estimator reports placebo effects for the six years before treatment, which play the role of a pre-trend test, and standard errors from 200 bootstrap replications. I apply it to the main panel: 188 treated and 109 never-treated commuting zones observed in every year from 2005 to 2024.

Appendix Figure A3 of the main paper and Panels B and C of Table OA6 report the estimates. The placebo estimates for the six years before treatment are all close to zero, so the synthetic comparison group tracks the treated commuting zones closely before Amazon’s entry. After entry, the estimates rise gradually, as in the [Callaway and Sant’Anna \(2021\)](#) results: the bachelor’s share increases by 1.24 percentage points six years after treatment and the master’s share by 0.69. The overall estimate is 0.67 percentage points for the bachelor’s share (standard error 0.17) and 0.45 for the master’s share (standard error 0.11); projecting out the log adult population gives 0.57 and 0.40. The shares with at least a high school diploma and at least some college do not rise.

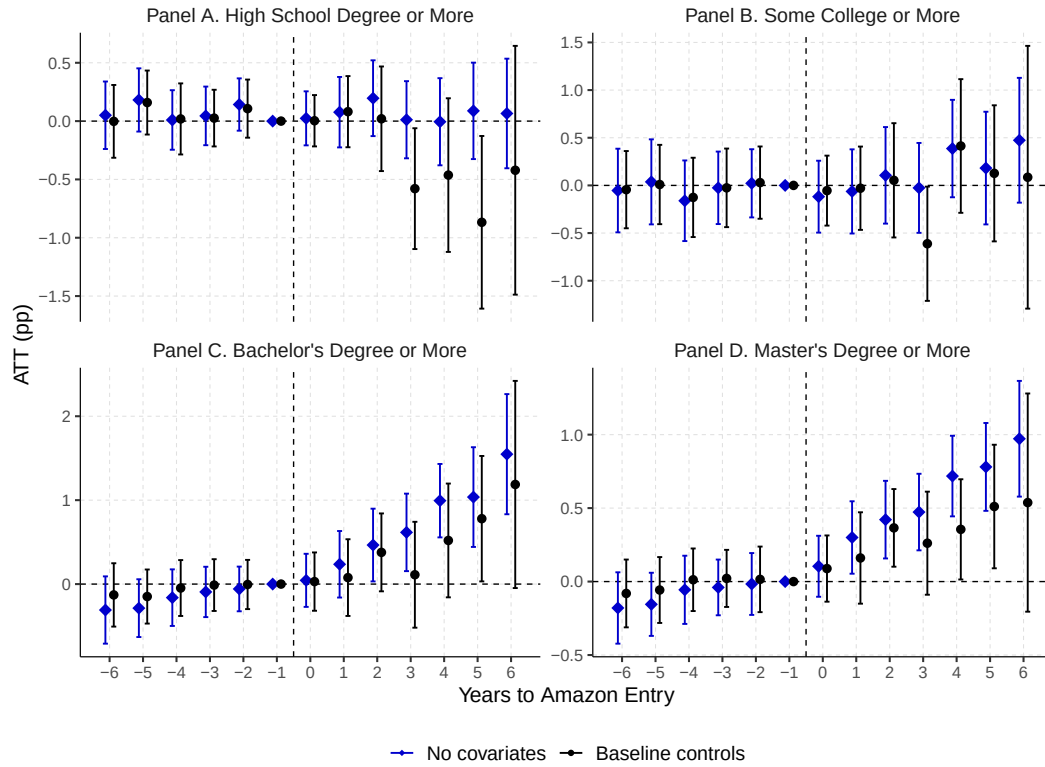
IPUMS microdata. The second data set is built from IPUMS ACS 1-year microdata, allocated from PUMAs to counties and commuting zones as described in Section 3.1. It covers all 577 commuting zones observed in every year from 2005 to 2024, including those without a county of 65,000 residents, of which 204 receive an Amazon facility. Table OA7 and Figure OA6 report the [Callaway and Sant’Anna \(2021\)](#) estimates, aggregated across treatment cohorts. The share of adults with a bachelor’s degree rises by 0.55 percentage points (standard error 0.20) and the share with a master’s degree by 0.45 percentage points (standard error 0.11); with baseline controls, the effects are 0.32 and 0.30. Without controls, the bachelor’s coefficients drift upward in the years before entry, reflecting that the largest and fastest-growing commuting zones, which Amazon entered first, were already becoming more educated; with controls, the pre-treatment coefficients are flat.

Table OA6: Effects of Amazon's Entry on Educational Attainment: ACS 1-Year Estimates, 2005–2024

	High School+ (1)	Some College+ (2)	Bachelor's+ (3)	Master's+ (4)
<i>Panel A. CSDID</i>				
Amazon Entry	–0.013 (0.138)	0.055 (0.167)	0.541*** (0.185)	0.422*** (0.117)
<i>Panel B. Synthetic DiD, no covariates</i>				
Amazon Entry	–0.251** (0.127)	0.101 (0.165)	0.675*** (0.168)	0.446*** (0.113)
<i>Panel C. Synthetic DiD, log adult population projected out</i>				
Amazon Entry	–0.180 (0.130)	0.102 (0.171)	0.566*** (0.176)	0.397*** (0.104)
Mean Dep. Var.	87.54	58.46	27.99	10.29
SD	4.59	7.14	6.60	3.28
Treated CZs	188	188	188	188
Never-treated donor CZs (SDID)	109	109	109	109
N (CSDID)	3,572	3,572	3,572	3,572

Notes: Dependent variable is the percentage-point share of adults aged 25 and older completing at least the specified schooling level, from published ACS 1-year estimates for counties with at least 65,000 residents (table B15002), 2005–2024, aggregated to 2020 commuting zones (2020 is not available). The sample is the balanced panel of commuting zones observed in every year. Panel A reports the doubly-robust staggered difference-in-differences estimator of [Callaway and Sant'Anna \(2021\)](#) without covariates, aggregated over all post-entry years (simple aggregation), in commuting zones that receive an Amazon facility between 2010 and 2024, with not-yet-treated commuting zones as comparisons. Panels B and C report the staggered synthetic difference-in-differences estimator of [Arkhangelsky et al. \(2021\)](#), with never-treated commuting zones as donors; Panel C projects out the log adult population. A commuting zone is treated from the year before its first facility opens. Standard errors in parentheses: multiplier bootstrap (1,000 draws) clustered at the commuting-zone level in Panel A; 200 bootstrap replications in Panels B and C. Mean Dep. Var. and SD are pre-treatment values in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure OA6: Event-Study Estimates With and Without Baseline Controls: IPUMS ACS Microdata, 2005–2024



Notes: Each panel plots event-time coefficients from two specifications: without covariates (blue diamonds) and conditioning on baseline controls (black circles), namely log average commuting-zone population in 2005–2009 and the change in log population between 2005–06 and 2008–09. Bars show uniform (simultaneous) 95% confidence bands from the multiplier bootstrap, clustered at the commuting-zone level. Event time $t = -1$ is normalized to zero and the vertical dashed line marks the treatment year, defined as the year before Amazon's first facility opens in the commuting zone (to allow for anticipation). Dependent variable is the percentage-point share of adults aged 25 and older completing at least the specified schooling level, from IPUMS ACS 1-year microdata (2005–2024) aggregated to 2020 commuting zones using person weights. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024. Estimates are the doubly-robust staggered difference-in-differences event-study coefficients of [Callaway and Sant'Anna \(2021\)](#), comparing treated commuting zones to not-yet-treated controls.

Table OA7: Effects of Amazon's Entry on Educational Attainment: IPUMS ACS Microdata

	High School+ (1)	Some College+ (2)	Bachelor's+ (3)	Master's+ (4)
<i>Panel A. No covariates</i>				
Amazon Entry	0.077 (0.129)	-0.012 (0.168)	0.548*** (0.201)	0.453*** (0.110)
<i>Panel B. Baseline controls</i>				
Amazon Entry	-0.425*** (0.141)	-0.263 (0.192)	0.320 (0.197)	0.300*** (0.109)
Mean Dep. Var.	86.79	55.74	25.33	9.13
SD	4.52	6.98	6.15	3.00
Commuting zones	204	204	204	204
N	4,080	4,080	4,080	4,080

Notes: Dependent variable is the percentage-point share of adults aged 25 and older completing at least the specified schooling level, from IPUMS ACS 1-year microdata (2005–2024) aggregated to 2020 commuting zones using person weights and population-weighted PUMA-to-county allocation factors. Estimates are the doubly-robust staggered difference-in-differences ATT of [Callaway and Sant'Anna \(2021\)](#), aggregated across treatment cohorts (group aggregation), comparing treated commuting zones to not-yet-treated controls. Sample is the balanced panel of commuting zones that receive an Amazon facility between 2010 and 2024; a commuting zone is treated from the year before its first facility opens (anticipation). Panel B conditions on baseline controls: log average population and the change in log population between 2005–06 and 2008–09. Standard errors from the multiplier bootstrap (1,000 draws), clustered at the commuting-zone level, in parentheses. Mean Dep. Var. and SD report the pre-treatment mean and standard deviation in treated commuting zones. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

References

- ARKHANGELSKY, D., S. ATHEY, D. A. HIRSHBERG, G. W. IMBENS, AND S. WAGER (2021): “Synthetic Difference-in-Differences,” *American Economic Review*, 111, 4088–4118.
- CALLAWAY, B. AND P. H. SANT’ANNA (2021): “Difference-in-Differences with Multiple Time Periods,” *Journal of Econometrics*, 225, 200–230.
- CHARLES, K. K., E. HURST, AND M. J. NOTOWIDIGDO (2018): “Housing Booms and Busts, Labor Market Opportunities, and College Attendance,” *American Economic Review*, 108, 2947–2994.